

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

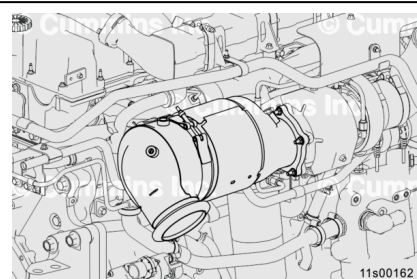
Additional Service Items

- Putty knife.

General Information

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

The aftertreatment three way catalyst elements contained in the aftertreatment system are made of brittle materials. Do not drop or strike the side of the aftertreatment as damage to the aftertreatment three way catalyst element can result.

Note : If the three way catalyst is being replaced due to progressive damage from engine oil or fuel, clean the exhaust tailpipe from the turbocharger outlet to the aftertreatment three way catalyst.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

⚠ WARNING ⚠

Wait for the aftertreatment system to cool down before working on the hot exhaust or exhaust components. Failure to do so could result in personal injury.

⚠ CAUTION ⚠

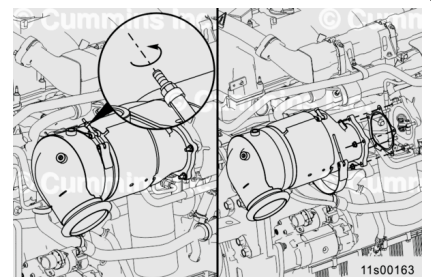
The aftertreatment primary three way catalyst can remain hot for a long time after the engine has stopped.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the exhaust pipe connector. See equipment manufacturer service information.
- Remove oxygen sensor from the primary three way catalyst. Refer to Procedure 019-100 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-100.html>)
- Remove temperature sensor from the primary three way catalyst. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-449.html>)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

The aftertreatment primary three way catalyst elements contained in the aftertreatment system are made of brittle materials. Do not drop or strike the side of the aftertreatment system as the damage will result.

Remove the clamp of the primary three way catalyst.

Remove the nuts from the primary three way catalyst.

Remove the primary three way catalyst from turbocharger and discard the gasket.

Clean and Inspect for Reuse

Note : If the primary three way catalyst is being replaced due to progressive damage from engine oil or fuel, clean the exhaust tailpipe from the turbocharger outlet to the aftertreatment primary three way catalyst.

- Remove any residual gasket material from the flanges on the primary three way catalyst with a putty knife.
- Inspect the mounting strap threads for damage.
- Replace the mounting strap if damage is found.

Note : Avoid dropping fragments of gasket into the primary three way catalyst.

Inspect the primary three way catalyst:

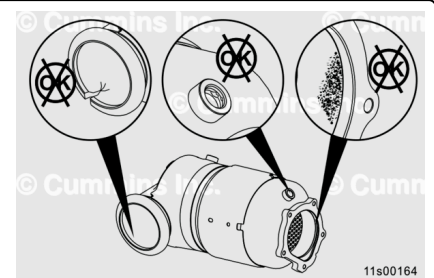
- Whether corrupt excessively
- Whether the primary three way catalyst is broken, such as the breach or the crack in the primary three way catalyst.

Replace the primary three way catalyst if necessary.

Note : If the primary three way catalyst is polluted by the coolant liquid, it must be replaced with a new one.

If the primary three way catalyst is replaced, the secondary three way catalyst **must** be replaced at the same time.

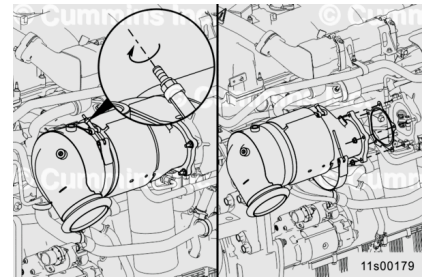
Check all the tube connector and the mounting bracket.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the primary three way catalyst to the turbocharger.

Tighten the nuts to the primary three way catalyst.

Torque Value: 23 n•m [204 in-lb]

Install the clamp of the primary three way catalyst.

Torque Value: 10 n•m [89 in-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

- Install oxygen sensor to the primary three way catalyst. Refer to Procedure 019-100 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-100.html>)
- Install temperature sensor to the primary three way catalyst. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-449.html>)
- Connect the exhaust pipe connector. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the vehicle on a dynamometer or perform a road test with the engine at rated load for a minimum of 5 minutes to be sure the aftertreatment system is performing properly.
- Check for fault codes and exhaust leaks.

